PRD: Rx Shortage Intelligence

Proactive signal surfacing for hospital drug shortages

Document type: Product Requirements Document (Prototype Scope) **Author:** Anton Verenitch **Date:** April 2026 **Status:** Working draft for v0.1 prototype build

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1. Product Summary

Rx Shortage Intelligence is an AI-assisted operational intelligence product for hospital pharmacy leadership and clinical pharmacists. It proactively surfaces drug shortage events affecting a hospital's formulary and active medication orders, classifies operational impact and clinical severity, recommends evidence-grounded therapeutic alternatives with citations, and routes outputs into the pharmacy workflow with human-in-the-loop controls.

The product transforms drug shortage response from a **reactive, fragmented, manual workflow** into a **proactive, structured, citation-backed daily briefing** — purpose-built for an industry in the middle of an unprecedented drug shortage crisis and underserved by legacy clinical reference tools that are perceived as slow, dense, and difficult to navigate.

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2. Why Now: The Drug Shortage Crisis is Reshaping Hospital Pharmacy Operations

2.1 The crisis is current, severe, and getting worse

- **75%+ of currently-active US drug shortages started in 2022 or later** — meaning the crisis is accelerating, not stabilizing.
- **Drug shortages cost US hospitals an estimated $359M/year in labor alone** (Vizient and other industry estimates), before patient safety costs, treatment delays, or readmission impact.
- **High-profile shortages in the past 24 months** — methotrexate IV (oncology), GLP-1 agonists (diabetes/obesity), stimulants (ADHD), sterile injectables, IV saline, cisplatin, carboplatin. Many remain unresolved at time of writing.
- **Hospital pharmacy directors report drug shortage management is the single most labor-intensive operational topic** in their week — surpassing P&T meeting prep, inventory management, and budget reporting (per ASHP and AHA surveys).

2.2 The market has recognized the gap, but solutions are incomplete

The industry has acknowledged that **clinical content and pharmacy inventory data need to be unified** for shortage response — partnerships between drug reference vendors and pharmacy inventory management systems (e.g., publicly announced integrations between drug-information products and inventory platforms like OrbitalRX) reflect this recognition.

But the available solutions remain **reactive lookup tools, not proactive signal surfacing systems.** They show inventory and clinical data side-by-side; they do not tell the pharmacy director what to act on this morning.

2.3 Legacy clinical reference tools are perceived as a bottleneck, not a solution

Documented and widely reported customer feedback on incumbent hospital drug reference tools — from KLAS reports, public software review platforms, and app store reviews — surfaces consistent themes:

> *"Layout could be improved, leading to difficulties in finding information quickly and efficiently."*

> *"Search functionality is clunky… it's best for broad single-term searching… finding specific information can be cumbersome."*

> *"Depth of information can be overwhelming."*

> *"Users need to know how to read and navigate the actual drug information in advance for the product to deliver value."*

> *"The UI/UX is absolutely terrible. Everything is hard to read and visually cluttered."*

These are not minor cosmetic gripes. They reflect three structural product problems for a category that is supposed to *speed up* clinical decision-making:

1. **Slow navigation** — clinicians wait for what should be instant
2. **The "clunkiness" factor** — multiple clicks, dense screens, unintuitive interaction patterns
3. **High learning curve** — the product requires training to deliver value

For a pharmacy director responding to an active shortage at 7:00 AM, "I have to know how to navigate the tool to get value out of it" is not acceptable. **Speed and intuitiveness are clinical safety features, not UX preferences.**

2.4 The opportunity

The combination of (a) a worsening shortage crisis, (b) legacy tools that are slow and clunky, and (c) the maturity of agentic AI for grounded clinical reasoning creates a clear opportunity:

> *Build the proactive, intuitive, fast shortage intelligence experience that legacy lookup tools were never designed to be — where AI does the cross-referencing the pharmacist used to do*

> *manually, the briefing is waiting when the user arrives, and trust is earned through citations, transparency, and human-in-the-loop controls.*

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3. Target Customer

3.1 Buyer / Relationship Owner

Persona: Karen Chen, PharmD, MBA — Director of Pharmacy, Memorial Health System (450-bed community-academic IDN)

Attribute	Description
Role	Director of Pharmacy / Chief Pharmacy Officer
Scope	Owns inpatient pharmacy operations, P&T governance, formulary management, clinical pharmacy services, drug shortage response
Reports to	CMO or COO
Team size	8-15 clinical pharmacists, 30-60 pharmacy techs
Budget authority	$50K-$500K for clinical decision support / pharmacy technology
Top OKRs	(1) Reduce medication-related adverse events; (2) Manage drug spend within ~5% of forecast; (3) Maintain Joint Commission and CMS compliance; (4) Operational efficiency of pharmacy team
Primary pain (current state)	Drug shortages: unpredictable, labor-intensive, safety-relevant. Pharmacist team burning out on manual cross-referencing.
What she's quietly tired of	Tools that are powerful but slow. Tools that require training to extract value. Tools that show data but don't drive action.

3.2 Daily User

Persona: Marcus Patel, PharmD — Clinical Pharmacist

Attribute	Description
Role	Inpatient clinical pharmacist; informal "shortage point of contact" by rotation
Daily activities	Order verification, drug interaction review, prescriber consultations on alternatives, P&T monograph drafting (rotating), shortage workarounds
Tools used today	EHR (Cerner/Epic), legacy drug reference, ASHP shortage list, hospital formulary spreadsheet, secondary drug reference, email, phone
Top frustration	"Every shortage means I'm checking 4-5 sources and answering the same prescriber phone calls over and over. None of these tools tell me what's *new* this morning vs. yesterday."
Trust requirements	Will not act on a recommendation without a citation. Will not trust a tool that has produced a hallucinated answer even once.

3.3 Stakeholders

- **P&T Committee** — receives shortage briefings, makes formulary decisions
- **Prescribing clinicians** — receive alternative recommendations at point-of-prescribing (downstream)
- **Pharmacy purchasing** — needs supply chain context for sourcing
- **Pharmacy informaticist** — owns clinical decision support configuration
- **Health system leadership** — wants reporting on shortage exposure and mitigation effectiveness

3.4 Segmentation Attributes

Attribute	Profile
Demographic	US health systems, 100+ beds, with active P&T committee and clinical pharmacy program
Psychographic	Evidence-driven, regulation-aware, conservative on patient safety, increasingly open to AI when it is grounded, transparent, and reviewable
Behavioral	Already a customer of one or more clinical reference platforms. Active formulary management. Has a pharmacy informatics function.
Trigger condition for buying	Active shortage event in the last 90 days that consumed significant pharmacist hours and/or caused a near-miss safety event

4. Customer Problems to be Solved

Stated where possible in the user's own voice (informed by KLAS reviews, ASHP literature, public app and software review platforms, and customer interview synthesis from secondary research).

Top problems

Problem 1: Reactive shortage discovery

> *"We find out a drug is in shortage when a nurse calls pharmacy because she can't get her dose dispensed. By then it's already a problem."*

Problem 2: Manual cross-referencing across fragmented sources

> *"To respond to one shortage, I'm checking the FDA shortage list, ASHP, our formulary spreadsheet, our drug reference tool for alternatives, and a second drug reference because the first one didn't show me what I needed. That's twenty minutes per drug, and we have multiple shortages weekly."*

Problem 3: No customer-specific impact assessment

> *"FDA tells me there's a shortage of meropenem. Great. How many of MY patients are on it right now? What's MY formulary's preferred alternative? Is THAT one available? I have to figure all of that out myself."*

Problem 4: Phone-call-based alternative recommendations

> *"Prescribers call me 30 times a day. 'Hey, can I substitute X for Y?' Each call is 5 minutes. That's 2.5 hours of my shift answering questions I've answered before."*

Problem 5: Slow and clunky reference tools

> *"The reference tool we have has the data, but it's not fast and it's not easy. I have to know exactly how to navigate it to get a useful answer. By the time I'm three clicks deep, the prescriber's already on a different question."*

Problem 6: Steep learning curve

> *"New pharmacy hires take weeks to get comfortable with our reference tools. The tools are powerful but they aren't intuitive. We need training modules just to make our drug reference usable."*

Problem 7: No audit trail or evidence chain

> *"When I tell a prescriber to use Drug B instead of Drug A, I need to be able to show why — the clinical equivalence, the formulary status, the supply rationale. Today that's tribal knowledge."*

What customers are *not* asking for

- More content depth (the existing tools have plenty)
- More features layered on top of existing UI
- More integrations with more dashboards

What customers *are* asking for

- **Speed** — answers in seconds, not minutes
- **Intuition** — usable on day one without training
- **Relevance** — show me what affects *my* hospital, not the universe of drug data
- **Action** — tell me what to do, not just what's true
- **Trust** — show me where the recommendation came from

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5. Product Principles

These principles directly translate the customer problems above into design constraints. They are non-negotiable for v0.1.

Principle 1: Proactive over reactive

The user opens the dashboard and the briefing is already there. The user does not "ask" the system. The system has already done the cross-referencing. The user does the clinical judgment.

Principle 2: Speed is a clinical safety feature

Time-to-insight should be measured in seconds, not minutes. Median briefing time-to-render: under 60 seconds. Median drill-down to source citation: one click.

Principle 3: Intuitive on day one

A clinical pharmacist who has never seen the product should be able to triage a morning briefing within five minutes of first login, without training. No learning curve is acceptable.

Principle 4: Customer-relevance over universal content

The system surfaces what affects *this hospital, this formulary, these patients* — not the universe of FDA data. Filtering precision is more important than retrieval recall on raw signals.

Principle 5: Citation-first trust

Every recommendation links to its source. Every source is verifiable. The agent's reasoning is visible on demand. There are no black-box outputs.

Principle 6: Human-in-the-loop, always

The agent recommends. The pharmacist decides. The system never auto-acts. Confidence scoring, approval gates, and audit logs are non-negotiable.

Principle 7: Honest about scope

Synthetic data is labeled as synthetic. Prototype boundaries are explicit. The system never claims a capability it doesn't have.

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6. Opportunity Sizing & Business Case

6.1 Market context

- US hospitals collectively spend ~$359M/year in labor on shortage response
- 75%+ of active shortages started in the last 24 months — the problem is intensifying
- Average mid-size hospital spends 8-12 pharmacist FTE-hours per week on shortage management during active events
- Documented patient safety events from substitution errors during shortages (ISMP, ASHP literature)

6.2 Customer-level economic case

For a representative 450-bed health system (Memorial Health profile):

Metric	Today (manual + legacy tools)	With Rx Shortage Intelligence	Annual impact
Pharmacist hours on shortage response	~10 hrs/week × 50 wks	~3 hrs/week	**~350 hrs/yr saved**
Pharmacist labor cost @ $75/hr loaded	~$37,500	~$11,250	**~$26K savings**
Time-to-respond (median)	4-8 hours from notification to action	<30 minutes with morning briefing	**~5x faster**
Substitution-related errors (estimated)	2-4/yr	<1/yr	**Reduced ADE risk**
New pharmacist time-to-productivity on shortage workflow	4-6 weeks (with legacy tools)	<1 day (intuitive UX)	**Reduced onboarding burden**
P&T meeting prep time per shortage decision	2-4 hrs	<30 min	**~75% reduction**

6.3 Strategic value

For a clinical content portfolio, Rx Shortage Intelligence:

- **Defends against churn** to competitor reference tools by closing the speed and intuitiveness gap
- **Demonstrates a credible AI-ready clinical knowledge layer** with a concrete, valuable product
- **Creates a sibling capability to Formulary Intelligence** using shared architecture (cost-efficient delivery)
- **Opens an MCP-distribution surface** that customer-side AI agents and broader healthcare AI ecosystem can consume

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7. Hypotheses & Experiments

7.1 Primary hypothesis

> *If pharmacy directors and clinical pharmacists are given a proactive AI-assisted morning*

> *briefing — surfacing only the shortages affecting their formulary, classified by severity with customer-specific impact, recommending grounded alternatives with citations, and accessible without training — then median time-to-shortage-response will decrease from 4-8 hours to under 30 minutes, with no degradation in clinical accuracy and no increase in substitution-related adverse events.*

7.2 Secondary hypotheses

- **H2 (Signal-to-noise):** Customer-context filtering reduces signal volume by ⩾80% vs raw FDA shortage feed, while maintaining 100% recall on formulary-impacting shortages.
- **H3 (Trust):** Citation-grounded recommendations achieve ⩾90% pharmacist acceptance vs <50% for ungrounded LLM outputs.
- **H4 (UX speed):** Median time-to-insight (login → triaged briefing) is under 5 minutes vs ~20 minutes with legacy lookup tools.
- **H5 (Day-one usability):** Pharmacists with no prior product training successfully triage a sample briefing within first session ⩾90% of the time.
- **H6 (Adoption):** Pilot users use the daily briefing as their primary shortage workflow within 30 days (⩾80% daily active use).
- **H7 (Cross-functional):** Briefings shared with P&T committees reduce monograph prep time for shortage-driven decisions by ⩾50%.

7.3 Experiments to validate (production roadmap, not prototype scope)

Experiment	Method	Success threshold
Pilot deployment	3-5 pilot hospitals, 90-day measurement	Time-to-response reduced ⩾3x; pharmacist NPS ⩾40
Continuous internal eval	Growing eval set with daily/weekly scoring	⩾90% clinical appropriateness; 100% citation accuracy; <2% hallucination
First-login usability test	New pharmacist users, no training, shadowed first session	⩾90% successfully triage briefing in <5 min
Comparative timed task	A/B vs. status quo (manual + legacy tools)	Statistically significant time savings, no safety degradation
Customer-side eval	Pharmacists rate briefings 1-5 over 30 days	Median rating ⩾4

8. Success Metrics & OKRs

Modeled on the four outcome categories most aligned with clinical content product success: time-to-decision, data quality, adoption, retention — plus safety, which is foundational.

8.1 Objective 1: Reduce time-to-shortage-response

KR	Target	Measurement
Median time from shortage notification to documented response	<30 min (vs 4-8 hr baseline)	Workflow timestamps
Pharmacist hours/week on shortage handling	<4 hrs (vs ~10 hrs baseline)	Self-reported time tracking
% of shortages addressed within same business day	⩾95% (vs ~50% baseline)	System logs

8.2 Objective 2: Maintain or improve clinical safety and quality

KR	Target	Measurement
Substitution-related adverse drug events	No increase from baseline	Hospital ADE reporting
Citation accuracy (every claim traceable to source)	100%	Automated eval
Clinical appropriateness of recommendations (eval set)	⩾90%	Human-rated eval
Hallucination rate	<2%	Automated + sampled human review
False negative rate (missed formulary-affecting shortage)	0%	Eval harness

8.3 Objective 3: Drive adoption and usability

KR	Target	Measurement
Daily active use among target users (90-day pilot)	⩾80%	Login analytics

KR	Target	Measurement
Time-to-first-value (login → triaged briefing, new user)	<5 min	First-session telemetry
Pharmacist acceptance rate of recommendations	⩾80%	UI accept/override capture
Pharmacist NPS	⩾40	Quarterly survey
Pilot-to-paid conversion	⩾60%	Sales pipeline

8.4 Objective 4: Drive retention and expansion

KR	Target	Measurement
Logo retention	⩾95% annually	Renewal tracking
Net Revenue Retention	⩾110%	ARR tracking
Cross-sell rate to adjacent formulary or content products	⩾40%	Customer mapping
Expansion to additional sites within IDN customers	⩾2 sites/customer year 2	Sales tracking

8.5 Operational efficiency metrics

Metric	Target
Median agent latency (briefing generation)	<60 sec
Median tokens per briefing	<8K input, <2K output
Cost per briefing (API spend)	<$0.05
API uptime (MCP servers)	⩾99.5%

8.6 Experience metrics by stakeholder

Stakeholder	Metric	Target
Pharmacy Director (buyer)	"Confidence in pharmacy team's shortage response" (1-5)	Median ⩾4
Clinical Pharmacist (user)	"Time saved per shortage event" (self-reported)	⩾75% reduction

Stakeholder	Metric	Target
Prescriber (downstream)	"Speed of alternative recommendation" (1-5)	Median ⩾4
P&T Committee	"Quality of evidence supporting shortage decisions" (1-5)	Median ⩾4
Patient (downstream)	Treatment delay from shortage	<2% of orders

9. Product Requirements & Functionality

9.1 Core functional requirements (v0.1 prototype)

FR-1: Proactive daily briefing generation

- System ingests current FDA Drug Shortage data (live)
- System ingests synthetic hospital formulary
- System ingests synthetic active orders snapshot
- System computes diff between current state and last-known state
- System filters changes to those affecting customer formulary or active orders
- System outputs a structured briefing organized by severity

FR-2: Severity classification

- Each surfaced item classified as Critical / Watch / Resolved
- Classification considers: formulary status, active order volume, alternative availability, route-of-administration constraints, therapeutic urgency
- Classification rationale exposed to user

FR-3: Therapeutic alternative recommendation

- For each Critical or Watch item, system retrieves clinically equivalent alternatives
- Alternatives ranked by: clinical equivalence, formulary status, current shortage status, route of administration match
- Each alternative includes cited rationale

FR-4: Citation and provenance

- Every claim links to source content
- Sources include: FDA Drug Shortage records, openFDA drug label sections, RxNorm relationships, hospital formulary records

- Citations clickable to source content

FR-5: Drill-down agent traces

- For any briefing item, user can view the agent's reasoning trace
- Trace shows: tool calls made, data retrieved, reasoning steps, classification logic
- Designed for pharmacist trust and audit defensibility

FR-6: HITL controls

- Each recommendation displays confidence score (high/medium/low)
- Low-confidence recommendations flagged for explicit pharmacist review
- High-confidence recommendations one-click acceptable
- All accept/override actions logged for audit and eval feedback

FR-7: Eval harness (internal)

- Eval set of 15 representative shortage scenarios
- Scoring across 5 dimensions: clinical appropriateness, citation accuracy, hallucination rate, severity accuracy, recall on formulary-affecting shortages
- Side-by-side comparison of prompt versions (v1 vs v2) demonstrating measured improvement

9.2 UX-specific requirements (in response to documented competitor weaknesses)

FR-8: Day-one usability

- New user can complete first briefing triage within 5 minutes of first login
- No required training, no required onboarding flow
- Empty states are self-explanatory

FR-9: Speed targets

- Dashboard loads with last briefing in <2 seconds
- Drill-down to citation in <1 second
- Agent rerun (drill-down agent trace) in <30 seconds

FR-10: Information density appropriate for power users

- Pharmacist scans, doesn't read
- Most important information visible without clicking

- Minimal modal interruptions

FR-11: Visual scan order

- Top: stats (today's date, items by severity, last run timestamp)
- Middle: severity-ordered briefing items (Critical → Watch → Resolved)
- Each item: visual severity badge → drug name → one-sentence summary → recommended action → expandable detail

9.3 Non-functional requirements

NFR-1: Latency — Briefing generation completes in <60 seconds end-to-end. **NFR-2: Cost** — Median briefing costs <$0.05 in API spend. **NFR-3: Reliability** — All public API failures handled gracefully with fallbacks. **NFR-4: Auditability** — Every agent action logged with timestamp, tool called, parameters, response, reasoning. **NFR-5: Honest scoping** — System never claims a capability it doesn't have. Synthetic data labeled as synthetic.

9.4 Out of scope for v0.1 (deferred to production roadmap)

- Background scheduler (briefings demonstrated via on-demand or pre-rendered execution)
- Push notification delivery (email, SMS, Slack, in-app)
- Real customer formulary integration
- Real EHR integration / CDS Hooks delivery
- Multi-tenancy and customer-specific deployment
- User authentication and role-based access
- Drug pricing intelligence (separate signal stream)
- Prior authorization workflow integration
- Pharmacy inventory data integration (would be added in production via partnership patterns established in the industry)

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10. User Experience Guidance

10.1 Desired UX

The product should feel like **a trusted briefing from a smart colleague**, not a search tool or a chatbot. The core UX principle: **proactive surfacing replaces self-directed querying.**

The user opens the dashboard and the briefing is already there. They scan, drill in where needed, and act. The system has done the work of cross-referencing; the user does the work of clinical

judgment.

This stands in deliberate contrast to legacy reference tools, which require the user to know what to search for, navigate to the right tab, click through the right hierarchy, and synthesize the answer themselves.

10.2 Critical user flows

Flow 1: Morning briefing review (primary flow)

1. Pharmacist opens dashboard at start of shift
2. Dashboard already shows: today's date, last briefing timestamp, summary stats (e.g., "3 critical, 2 watch, 1 resolved")
3. Briefing items rendered in severity order
4. Each item shows: drug name, severity badge, one-sentence summary, recommended action
5. Click expands to: full agent reasoning, citations, alternatives with comparison, accept/override
6. **Total time to triage all items: target <5 minutes**

Flow 2: Drill-down for high-stakes review

1. From briefing item, click "Show agent reasoning"
2. See ordered tool calls (inputs, outputs)
3. See severity classification logic
4. See alternative ranking rationale
5. Each citation clickable to source content
6. Accept, override, or escalate with reason

Flow 3: Eval review (internal/admin)

1. Open eval tab
2. View eval set with v1 vs v2 prompt scores side-by-side
3. Click any test case for input, expected output, actual output, score breakdown
4. Visualize score deltas across versions

10.3 Visual design aesthetic

- **Clinical-professional, not consumer-flashy**
- **Reduced color palette:** primary neutrals; reserved use of red (critical), amber (watch), green (resolved); no gradients

- **Information density appropriate for power users** — pharmacists scan, don't read
- **Typography:** sans-serif system stack, monospace for codes (NDC, RxCUI), high contrast
- **Citation-first** — citations visible by default, not hidden in tooltips
- **Tool call traces:** rendered as a stepped log, like a developer console — the transparency itself is part of the trust UX
- **Empty states:** "No shortages affecting your formulary today" reads as good news, with clear last-run timestamp
- **Loading states:** when the agent runs, show actual tool calls streaming in — transparency *is* the experience
- **No "AI sparkle" iconography** — this is clinical software

10.4 Anti-patterns (explicit rejections of legacy tool problems)

These design choices are *deliberate inversions* of documented competitor failures:

Legacy weakness	Anti-pattern we reject	Our design choice
"Layout is hard to navigate"	Multi-tab hierarchies that hide information	Single-screen scannable dashboard
"Search is clunky for specific information"	Search-first, browse-second design	Briefing-first; users don't need to search
"Information depth is overwhelming"	Showing everything, letting user filter	Showing only what affects this hospital
"Need to know how to use it for value"	Feature-first navigation that requires training	Outcome-first surface that explains itself
"Multi-click drill-downs"	Modal stacks and nested menus	Inline expansion; max 1 click to source
Conversational chatbot interface	Open-ended Q&A	Structured briefing; pharmacist scans, not types

10.5 Animation and motion

Restrained. Loading skeletons over spinners. Tool call traces stream in but don't bounce or animate. Severity color transitions are instant, not animated. Speed perception is the priority.

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11. Data Model

11.1 Core entities

Drug

- `rxcui`, `name`, `ndc_codes[]`, `route_of_administration`, `therapeutic_class`, `label_url`, `is_iv_only`

ShortageRecord

- `record_id`, `rxcui`, `status`, `reason_for_shortage`, `start_date`, `est_resolution_date`, `affected_presentations[]`, `last_updated`, `source_url`

Formulary (synthetic)

- `customer_id`, `drugs[]` with formulary_status, restrictions, preferred_alternatives

FormularyEntry

- `rxcui`, `formulary_status`, `restriction_criteria`, `preferred_alternatives[]`, `last_pt_review_date`

ActiveOrders (synthetic)

- `customer_id`, `orders[]` with rxcui, count, last_30_days, departments

BriefingRun

- `run_id`, `run_timestamp`, `customer_id`, `prompt_version`, `items_reviewed`, `items_surfaced`, `tool_calls[]`, `total_tokens_used`, `total_cost_usd`, `latency_ms`

BriefingItem

- `item_id`, `run_id`, `rxcui`, `severity`, `summary`, `rationale`, `alternatives[]`, `citations[]`, `confidence`, `recommended_action`, `user_action`, `user_action_timestamp`

EvalCase

- `case_id`, `input`, `expected`, `actual_v1`, `actual_v2`, `scores`

11.2 Data sources

Source	Type	Status	Notes
FDA Drug Shortage API	Real public	Live fetch	Authoritative shortage data
openFDA Drug Label API	Real public	Live fetch	Drug labeling, indications, contraindications, alternatives info
RxNorm API (NLM)	Real public	Live fetch	Drug normalization and therapeutic equivalence relationships
Hospital Formulary	Synthetic	Generated	30 representative drugs with realistic statuses
Active Orders	Synthetic	Generated	Volume signals for severity classification
Eval set	Hand-curated	Generated	15 scenarios spanning critical/watch/resolved cases

12. Tech Stack & Components

12.1 Components

Layer	Tool	Why
LLM	Anthropic Claude (claude-sonnet-4-6)	High-quality reasoning, native tool use, reliable structured output
Agent framework	Anthropic native tool use loop	No abstraction overhead; direct demonstration of agent loop
MCP	Custom MCP servers in Python	Demonstrates the architectural pattern
RAG	Lightweight retrieval over openFDA label JSON (chunk + simple keyword retrieval for v0.1)	Grounded clinical context
Public APIs	Direct HTTP to FDA Drug Shortage, openFDA, RxNorm	No-key public APIs, reliable, citable

Layer	Tool	Why
UI	Streamlit (single-file app)	Fast to build, professional enough, runs locally
Persistence	Local JSON files	Sufficient for prototype
Eval harness	Custom Python module with structured scoring	Demonstrates measurement discipline
Code repo	GitHub	Demonstrable artifact

12.2 Architectural pattern

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│                        STREAMLIT UI                          │
│           (Dashboard + Drill-down + Eval results)            │
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│                     AGENT ORCHESTRATOR                       │
│          (Anthropic Claude + tool use loop + RAG)            │
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12.3 MCP servers

- `fda_shortage_server` — `get_current_shortages()`, `get_shortage_detail()`. Wraps FDA Drug Shortage API.

- `drug_label_server` — `get_drug_label_sections(rxcui, sections)`, `search_labels_by_indication()`. Wraps openFDA.
- `rxnorm_server` — `normalize_drug_name(name)`, `get_therapeutic_alternatives(rxcui)`. Wraps RxNorm.

12.4 Why these choices

- **Claude over alternatives:** native MCP support, strong tool use, Anthropic API
- **MCP over function-calling:** demonstrates future-state distribution architecture
- **Public APIs over synthetic-only data:** real, citable, defensible
- **Streamlit over React:** ship in hours, not days
- **Local JSON over database:** no infra, demoable
- **Native agent loop over abstraction libraries:** transparency over abstraction

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13. Human-in-the-Loop & Governance Controls

13.1 Confidence-based routing

Confidence	Behavior
High (≥85%)	Surfaced as recommendation; one-click accept
Medium (60-85%)	Surfaced with "review recommended" badge
Low (<60%)	Surfaced with "manual review required"; no one-click accept

13.2 Hard stops

The agent never auto-acts. Every recommendation requires explicit pharmacist accept, override, or escalate.

13.3 Audit trail

Every briefing run captures: tools called, reasoning chain, classification logic, citations, user actions with timestamps.

13.4 Editorial governance

Clinical content team defines: allowed sources for citations, severity classification rules, therapeutic equivalence criteria, sampling-review cadence. PM and content team co-author AI feature requirements.

13.5 Failure modes

Failure mode	Mitigation
Hallucinated drug or alternative	RAG grounding + citation requirement; <2% target
Missed formulary-affecting shortage	Recall test; 100% required
Over-flagging	Customer-context filtering; severity gating
Stale data	Live API fetch + last-known cache fallback
API outage	Graceful degradation, surface what's available

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14. Roadmap from Prototype to Production

Phase	Scope
v0.1 (this prototype)	On-demand briefing, synthetic formulary, single user
v0.2	Real customer formulary integration, multi-user
v0.3	Background scheduler + email/Slack push, customer configuration
v0.4	EHR CDS Hooks integration: alerts at point-of-prescribing
v0.5	Watchlist subscription, role-specific briefings
v1.0	MCP server exposure for customer-side AI agent consumption
v1.x	Pharmacy inventory data integration; outcome correlation analytics

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15. Risk Register

Risk	Likelihood	Impact	Mitigation
Hallucination causes patient safety event	Low	Critical	RAG + HITL + eval harness + clinical content review

Risk	Likelihood	Impact	Mitigation
Pharmacist trust adoption gap	Medium	High	Citation-first UX; prompt iteration; pilot-and-learn
Competitive response from incumbents	High	Medium	Move fast on speed, intuitiveness, MCP, EHR integration
Over-flagging causes alert fatigue	Medium	High	Customer-context filtering; severity gating
Editorial governance bottleneck	Medium	Medium	Tight partnership with content team; co-authored requirements
FDA shortage data quality varies	Medium	Medium	Cross-reference with ASHP in production; flag uncertainty

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16. Open Questions

These would be validated in the first 30-60 days in role:

1. What's the current state of the formulary data model — what's structured, what's free text?
2. Is there an existing EHR partnership that would accelerate CDS Hooks integration?
3. How is the clinical content team currently structured around shortage and formulary content?
4. What's the customer demand split between pharmacy directors and P&T committees?
5. Is there an existing pricing intelligence data layer that could plug into shortage decisions?
6. How is the org thinking about MCP server exposure as a distribution strategy?
7. What does existing user research tell us about pharmacy director morning workflow?

End of PRD v0.2